

Assignment 1: Network Latencies, Ping & Traceroute

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Repository: <https://github.com/bvin02/CS42200-Fall26-Assignment1>

The experiment completed the full ping list and five traceroutes.

Measurement location: Purdue University, West Lafayette, IN. Public IP: 173.230.6.130.

Coordinates used for distance: 40.42370, -86.92120.

I used the server list from <https://iperf3serverlist.net/api/servers>, sent 5 pings per IP, and used 3 probes per traceroute hop. Seed 422 controls the random traceroute selection.

Rerun command: `./run.sh --origin-label "Purdue University, West Lafayette, IN" --origin-lat 40.4237 --origin-lon -86.9212 --repo-url https://github.com/bvin02/CS42200-Fall26-Assignment1`

Code sections: `experiment.py:78` (fetch); `experiment.py:129` (ping_ip); `experiment.py:154` (geolocate); `experiment.py:147` (haversine); `experiment.py:193` (trace_ip); `experiment.py:168` (parse_trace); `experiment.py:325` (render). Raw command output is in `results/raw`.

1. Ping results and distance

I tested 188 unique IP addresses, including my public IP. 184 responded and 4 did not. The plot contains the 184 addresses with both RTT and coordinates. Missing replies are not treated as zero.

My public IP averaged 11.194 ms and is shown at zero distance as a local reference.

The plot compares great-circle distance with minimum, average, and maximum RTT. The Pearson correlation between distance and average RTT was 0.929, excluding my own IP.

The nearest geolocated responding server was 185.93.1.65 at 170 km with 23.25 ms mean RTT; the farthest was 157.66.210.199 at 15971 km with 288.09 ms. The largest observed min-to-max spread was 250.75 ms at 96.45.44.87 (208.70 to 459.45 ms).

RTT usually increased with distance, but distance was not the only factor. Internet routes are not straight lines, and processing, link speed, and queueing also add delay.

Minimum RTT is the best baseline seen during the test. Maximum RTT includes the worst delay seen. A large min-max gap shows that network conditions changed between packets.

Coordinates came from ipwho.is and are approximate. The full data is in the appendix and results/ping.csv.

2. Traceroute results and hop count

I shuffled the IP list with seed 422 and tried addresses until five destinations responded. 5 paths were attempted and 5 completed.

41.210.185.162: 19 hops, 260.99 ms; 212.47.219.73: 17 hops, 134.28 ms; 143.244.63.144: 12 hops, 212.70 ms; 143.14.22.69: 21 hops, 124.10 ms; 69.168.185.43: 10 hops, 46.02 ms

Asterisks are unanswered hops. A trace only counts when the final destination responds. Hop count is the final TTL, including unanswered intermediate hops.

The stacked chart subtracts each responding hop RTT from the next responding hop RTT. Hatched bars cross unanswered hops. The black diamond shows the destination RTT.

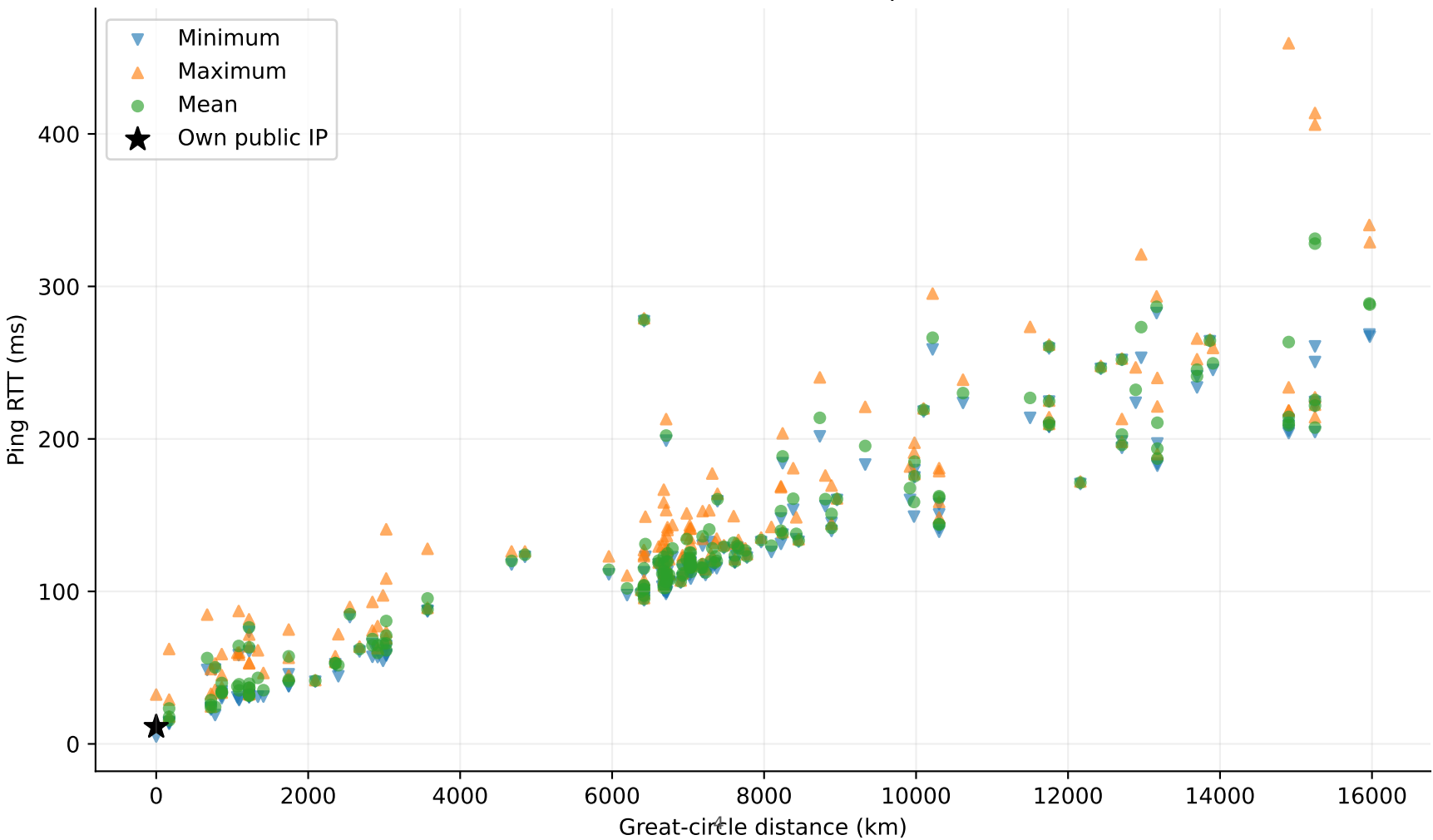
There were 22 negative differences. Each traceroute probe is a different packet, so queueing and return paths can vary. A negative bar is measurement variation, not negative link delay.

The correlation between hop count and final RTT was 0.365. There are only five paths, so this result only describes this run.

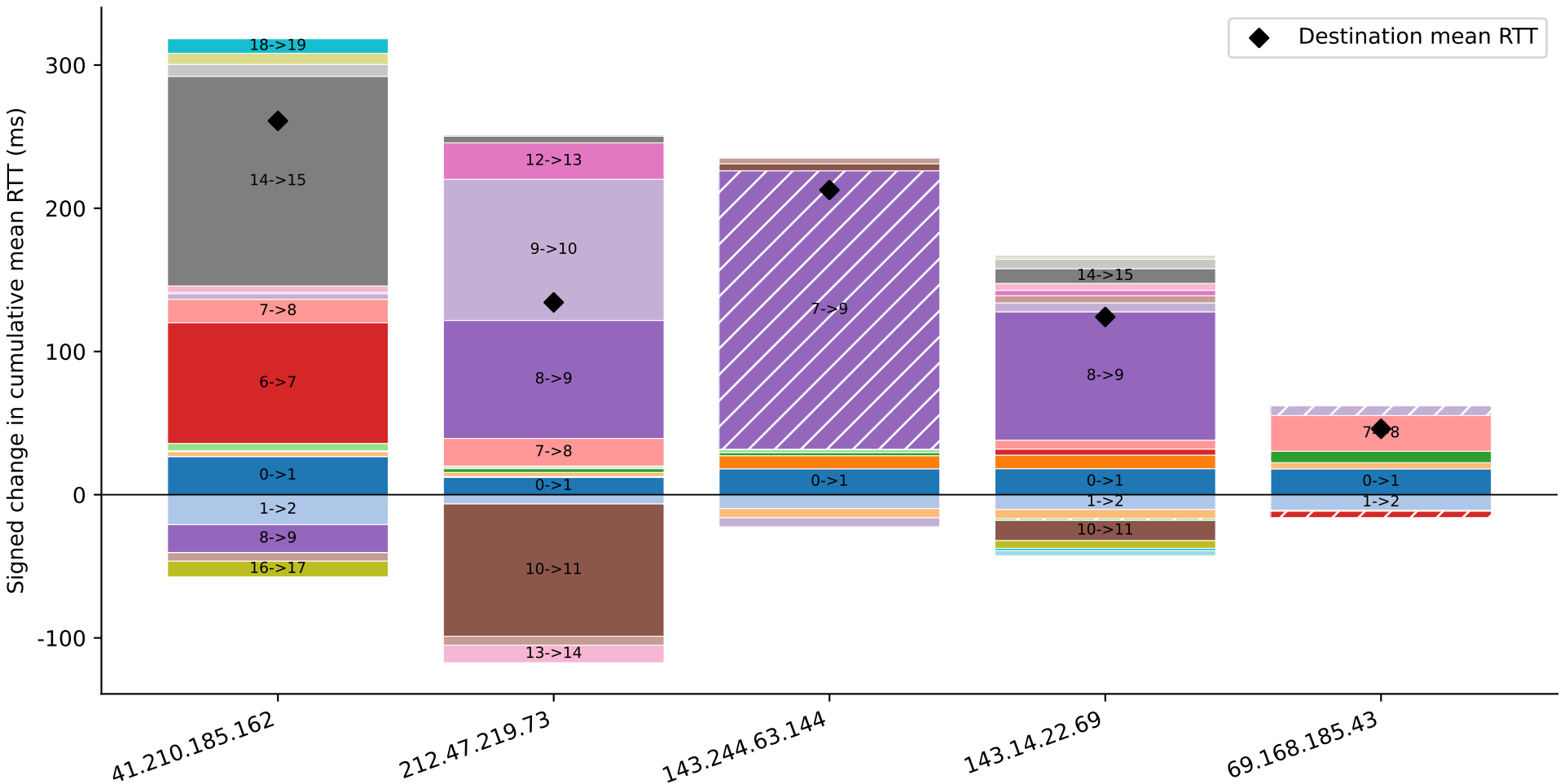
The largest destination TTL was 21 for 143.14.22.69, with 124.10 ms RTT. The highest RTT was 260.99 ms for 41.210.185.162 at 19 hops. These are different paths: more hops did not imply the highest RTT in this sample.

More hops can add delay, but hop count alone does not predict RTT. A few long links can take more time than many short links. Route length and queueing also matter.

Distance versus round-trip time

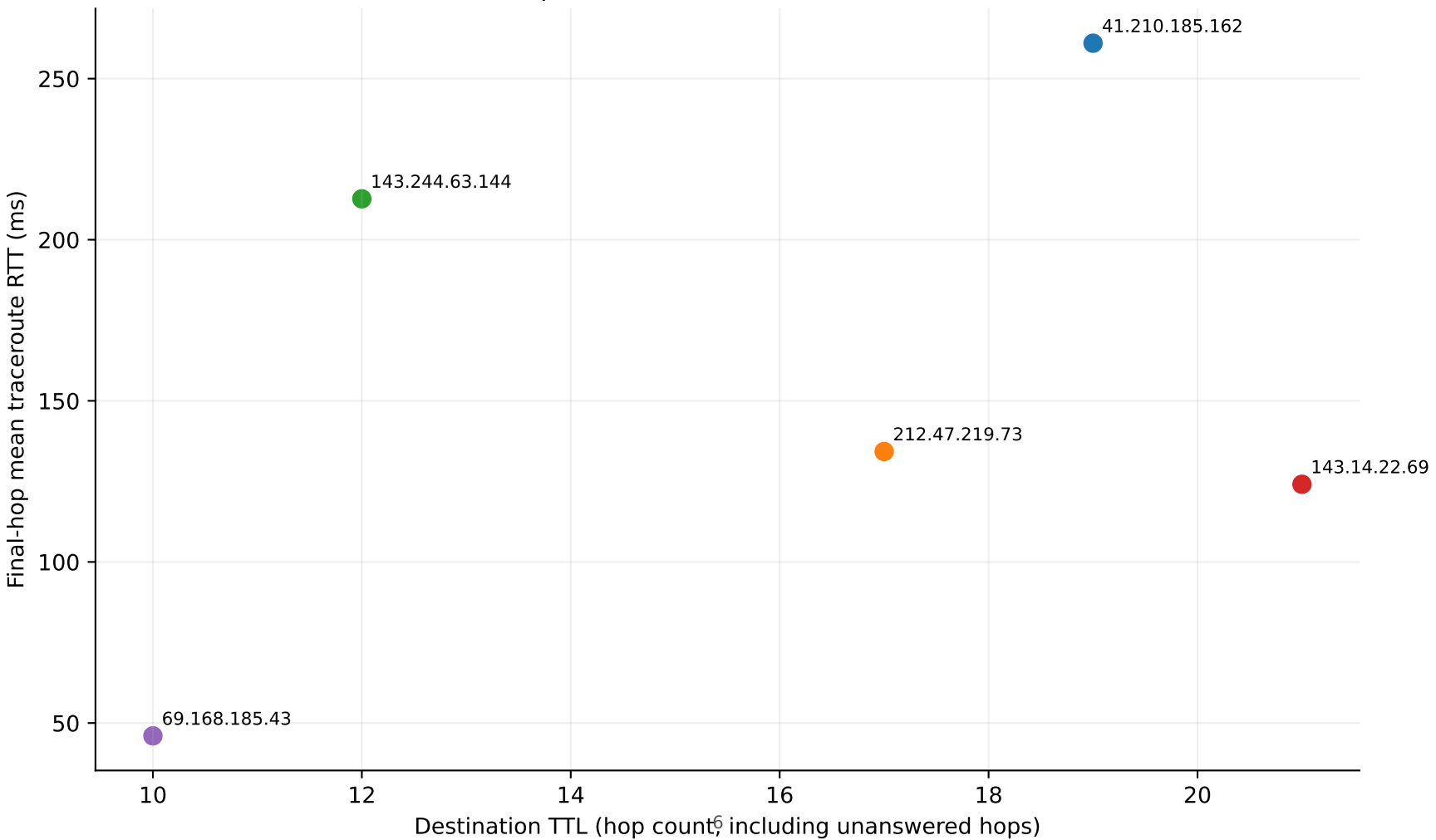


Traceroute latency breakdown: adjacent responding TTLs



Segments labeled previous TTL -> current TTL. Hatched = spans unanswered hops.
 Negative changes retained; these are RTT differences, not measured link delays.

Hop count versus destination RTT



Ping results and coordinates - 1-28

IP	Status	Lat	Lon	km	Min ms	Mean ms	Max ms
101.98.9.197	responsive	-36.85	174.76	13175.24	182.45	193.64	221.28
101.98.9.201	responsive	-36.85	174.76	13175.24	197.00	210.61	240.00
102.214.66.19	responsive	12.37	-1.53	8733.40	201.68	213.82	240.34
102.214.66.39	responsive	5.56	-0.20	9330.47	183.20	195.39	221.02
102.215.35.132	responsive	-1.28	36.82	12890.48	223.74	232.19	246.90
103.146.200.98	responsive	-9.48	147.15	13697.68	241.07	245.42	252.25
103.214.44.130	responsive	53.48	-2.24	6196.89	97.58	102.00	110.43
104.143.1.141	responsive	48.21	16.37	7614.79	118.93	123.56	131.53
105.235.237.2	responsive	3.76	8.78	10217.90	258.70	266.37	295.31
108.62.52.129	responsive	40.71	-74.01	1090.40	29.29	39.10	60.01
109.61.86.65	responsive	42.36	-71.06	1338.50	30.94	43.30	61.39
109.61.94.65	responsive	38.73	-9.15	6437.82	122.72	131.05	149.01
121.127.43.65	responsive	20.59	-100.39	2548.17	83.17	85.06	89.80
138.199.14.66	responsive	43.30	5.38	7229.03	110.80	112.46	114.58
138.199.4.1	responsive	-23.55	-46.64	8243.85	137.27	137.99	138.91
138.199.57.129	responsive	43.71	-79.40	719.90	23.53	23.93	24.54
139.60.160.88	non-responsive	40.71	-74.01	1090.40	-	-	-
141.98.136.66	responsive	50.12	8.68	7034.59	113.20	115.73	120.23
143.14.22.69	responsive	58.51	13.16	6793.80	122.41	128.20	143.54
143.244.63.144	responsive	-33.87	151.21	14904.22	203.80	210.66	233.81
145.0.6.51	responsive	52.09	5.12	6708.96	98.35	102.32	112.93
148.187.129.10	responsive	46.01	8.96	7299.69	116.70	119.72	124.49
148.230.45.213	responsive	40.42	-3.70	6753.47	105.74	107.59	111.35
148.230.59.38	responsive	37.34	-121.89	3026.80	64.29	65.88	69.30
148.230.60.200	responsive	-23.55	-46.64	8243.85	184.09	188.56	203.65
15.235.182.181	responsive	1.29	103.85	15249.81	224.25	225.58	227.29
154.81.51.4	responsive	-9.48	147.15	13697.68	233.76	241.10	265.82
156.146.40.65	responsive	48.15	17.11	7662.12	126.51	128.33	133.72

Ping results and coordinates - 29-56

IP	Status	Lat	Lon	km	Min ms	Mean ms	Max ms
156.146.52.1	responsive	41.01	28.95	8887.60	139.67	141.68	144.34
156.146.53.53	responsive	37.34	-121.89	3026.80	57.95	61.37	73.43
157.250.201.61	responsive	40.79	-74.06	1085.81	58.20	64.22	87.15
157.66.210.199	responsive	-6.21	106.85	15971.46	267.03	288.09	329.00
160.242.19.254	responsive	-8.84	13.23	11500.84	213.81	226.89	273.46
163.172.208.7	responsive	52.37	4.89	6679.93	106.72	122.72	166.72
169.150.202.193	responsive	32.08	34.78	9972.93	149.05	158.63	190.97
169.150.242.129	responsive	45.81	15.98	7753.01	124.73	126.88	128.20
169.150.252.2	responsive	37.98	23.73	8806.02	155.96	160.51	176.02
170.83.216.19	responsive	-1.21	-80.37	4676.68	117.66	120.01	126.22
173.243.131.29	responsive	45.41	-75.70	1067.56	30.61	37.74	59.73
176.23.128.102	responsive	1.29	103.85	15249.81	204.50	207.44	214.20
178.162.166.76	responsive	52.37	4.89	6679.93	105.17	107.09	108.17
178.162.226.176	responsive	50.12	8.68	7034.59	114.44	125.35	141.29
178.215.228.109	responsive	50.12	8.68	7034.59	108.38	112.74	121.81
178.217.143.20	responsive	50.05	21.41	7774.59	122.17	122.97	124.41
185.102.217.170	responsive	44.43	26.11	8454.89	132.59	133.29	134.21
185.102.218.1	responsive	52.37	4.89	6679.93	100.84	103.55	112.39
185.102.219.93	responsive	50.12	8.68	7034.59	114.73	115.65	116.64
185.118.248.162	responsive	55.68	12.57	6927.99	114.32	117.64	123.77
185.125.168.212	responsive	59.13	10.22	6612.35	115.69	120.23	129.44
185.15.210.116	responsive	52.52	5.72	6721.04	99.81	107.85	127.25
185.152.44.54	responsive	41.56	14.67	7960.95	132.36	133.63	135.30
185.152.65.113	responsive	50.09	14.42	7376.66	115.19	119.98	134.58
185.152.66.67	responsive	33.75	-84.39	775.41	48.81	50.19	52.77
185.152.67.2	responsive	34.05	-118.24	2845.64	57.11	64.65	93.08
185.180.12.40	responsive	48.21	16.37	7614.79	119.11	119.63	121.10
185.182.195.76	responsive	52.00	4.23	6660.25	103.15	111.61	129.68

Ping results and coordinates - 57-84

IP	Status	Lat	Lon	km	Min ms	Mean ms	Max ms
185.245.99.75	responsive	50.12	8.68	7034.59	112.92	116.88	130.99
185.246.208.67	responsive	52.23	21.01	7599.73	126.92	131.93	149.36
185.3.160.57	responsive	50.85	4.35	6728.87	120.36	124.92	142.04
185.59.221.51	responsive	51.51	-0.13	6418.48	112.56	115.12	123.02
185.59.223.8	responsive	40.71	-74.01	1090.40	28.72	35.13	58.49
185.76.9.135	responsive	59.33	18.07	6981.69	121.56	134.33	151.18
185.84.224.68	non-responsive	50.12	8.68	7034.59	-	-	-
185.93.1.65	responsive	41.85	-87.65	169.94	13.13	23.25	62.26
185.93.2.193	responsive	48.85	2.35	6710.83	99.71	106.38	123.70
185.93.3.50	responsive	40.42	-3.70	6753.47	106.36	110.71	121.17
189.25.101.151	responsive	25.08	55.31	11750.75	224.51	224.72	225.13
189.25.97.217	responsive	25.08	55.31	11750.75	208.13	209.10	209.83
189.25.98.63	responsive	25.05	121.53	12162.23	170.41	171.35	172.07
193.106.164.10	responsive	55.68	12.57	6927.99	108.82	111.74	122.50
193.106.166.153	responsive	55.68	12.57	6927.99	108.50	110.42	111.87
193.239.236.131	responsive	41.85	-87.65	169.94	13.87	17.75	29.05
194.127.172.176	responsive	52.37	4.89	6679.93	108.64	111.62	117.83
194.135.46.48	responsive	52.52	5.72	6721.04	106.24	112.46	136.44
194.24.162.86	responsive	51.51	-0.09	6420.39	94.14	95.03	95.60
195.162.68.38	responsive	64.14	-21.90	4852.50	122.69	124.08	126.22
195.181.162.195	responsive	25.77	-80.19	1743.89	38.04	42.14	56.47
195.69.189.215	responsive	41.26	69.22	10617.83	223.61	230.05	238.83
195.95.177.10	responsive	52.37	4.89	6679.93	107.73	113.46	134.10
197.227.12.18	responsive	-20.16	57.50	15965.58	268.46	288.85	340.26
197.227.5.218	responsive	51.51	-0.09	6420.39	277.20	278.07	279.09
198.60.22.18	responsive	40.11	-111.67	2093.37	40.81	41.45	41.83
2.56.188.136	responsive	32.78	-96.81	1223.42	32.27	39.49	63.99
201.225.207.2	responsive	8.99	-79.52	3570.68	87.36	88.22	89.23

Ping results and coordinates - 85-112

IP	Status	Lat	Lon	km	Min ms	Mean ms	Max ms
201.225.207.3	responsive	8.99	-79.52	3570.68	86.88	95.41	127.97
202.22.159.9	responsive	-22.27	166.45	12962.73	253.20	273.30	321.00
204.155.243.172	responsive	43.71	-79.40	719.90	22.53	28.77	49.06
207.211.214.65	responsive	50.85	4.35	6728.87	113.23	119.63	140.27
208.110.65.2	responsive	39.10	-94.58	670.64	48.52	56.30	84.80
209.40.123.215	responsive	32.78	-96.81	1223.42	31.61	36.71	53.03
209.58.159.68	responsive	32.78	-96.81	1223.42	30.61	31.50	32.60
212.237.217.41	non-responsive	60.17	24.94	7235.26	-	-	-
212.47.219.73	responsive	59.44	24.75	7278.47	132.38	140.62	153.21
212.60.92.134	responsive	13.45	-16.58	7386.42	159.13	160.49	164.02
212.76.127.243	responsive	32.06	34.86	9979.20	179.51	185.11	197.52
212.76.127.244	responsive	32.06	34.86	9979.20	174.83	175.51	176.93
213.158.175.240	responsive	30.06	31.25	9919.76	160.02	167.76	181.88
213.209.106.95	responsive	53.70	9.99	6902.19	105.77	106.46	107.16
216.232.132.95	non-responsive	48.45	-123.30	2991.30	-	-	-
217.18.95.142	responsive	52.37	4.89	6679.93	100.55	102.22	103.89
217.61.40.96	responsive	45.71	9.59	7359.08	118.43	122.86	131.96
23.105.41.222	responsive	32.78	-96.81	1223.42	60.69	63.32	71.69
23.105.9.80	responsive	38.90	-77.04	862.59	32.86	33.28	33.64
23.106.226.75	responsive	35.69	139.69	10303.47	150.52	161.55	178.82
23.106.244.70	responsive	-33.87	151.21	14904.22	212.59	214.42	218.55
23.106.58.162	responsive	51.51	-0.13	6418.48	98.81	104.10	124.11
23.108.99.54	responsive	1.29	103.85	15249.81	250.42	328.08	413.74
23.19.208.9	responsive	41.85	-87.65	169.94	14.23	15.47	17.89
23.249.54.234	responsive	25.77	-80.19	1743.89	37.77	41.08	44.78
23.249.55.42	responsive	25.08	55.31	11750.75	207.88	210.73	214.23
23.249.58.14	responsive	22.28	114.17	12708.84	198.34	202.89	213.07
23.249.60.154	responsive	35.69	139.69	10303.47	139.11	143.79	158.71

Ping results and coordinates - 113-140

IP	Status	Lat	Lon	km	Min ms	Mean ms	Max ms
23.81.166.99	responsive	37.34	-121.89	3026.80	58.63	71.07	108.45
23.81.207.31	responsive	33.45	-112.07	2356.56	50.94	52.83	57.48
23.82.138.78	responsive	25.77	-80.19	1743.89	45.67	57.42	75.00
23.82.193.66	responsive	47.61	-122.33	2914.04	61.13	64.61	77.24
27.123.215.194	responsive	47.91	106.88	10099.11	217.94	219.09	219.99
31.132.7.130	responsive	51.51	-0.13	6418.48	100.64	101.62	104.14
31.14.130.68	responsive	43.46	11.88	7651.09	128.70	129.70	131.36
37.19.203.1	responsive	42.70	23.32	8423.28	134.20	137.76	148.59
37.19.206.20	responsive	38.90	-77.04	862.59	34.44	39.94	58.89
37.19.216.1	responsive	29.76	-95.36	1410.85	31.03	35.24	46.52
37.19.218.65	responsive	50.45	30.52	8221.87	131.16	139.81	168.92
41.110.39.130	responsive	36.73	3.09	7470.89	128.41	129.27	130.29
41.210.185.162	responsive	0.32	32.58	12431.29	245.95	246.68	247.86
45.147.210.189	responsive	48.85	2.35	6710.83	99.54	106.32	128.62
45.250.25.133	responsive	38.90	-77.04	862.59	29.98	34.47	45.09
45.86.208.131	responsive	33.45	-112.07	2356.56	52.30	52.98	53.76
45.88.200.26	responsive	59.13	10.22	6612.35	117.69	118.50	118.98
49.205.75.2	responsive	12.97	77.59	13866.63	264.06	264.50	265.20
5.180.211.133	responsive	51.51	-0.09	6420.39	96.00	97.29	98.46
51.158.1.19	responsive	48.85	2.35	6710.83	198.84	202.21	212.99
51.158.1.21	responsive	48.85	2.35	6710.83	109.54	112.03	116.35
51.161.209.134	responsive	-33.87	151.21	14904.22	205.44	209.05	218.62
62.106.94.245	responsive	40.20	29.06	8959.39	159.96	160.54	160.98
62.176.224.217	responsive	52.42	10.78	7021.68	114.76	120.83	143.25
62.176.246.102	responsive	52.42	10.78	7021.68	111.02	121.85	141.06
62.176.246.171	responsive	52.42	10.78	7021.68	115.38	116.15	117.34
62.176.246.197	responsive	52.42	10.78	7021.68	117.36	118.30	119.11
62.176.246.39	responsive	52.42	10.78	7021.68	121.62	125.53	135.02

Ping results and coordinates - 141-168

IP	Status	Lat	Lon	km	Min ms	Mean ms	Max ms
62.176.246.73	responsive	52.42	10.78	7021.68	117.67	118.74	119.87
62.176.246.8	responsive	52.42	10.78	7021.68	114.92	117.60	124.87
62.176.247.4	responsive	52.42	10.78	7021.68	113.94	115.79	121.66
64.120.117.34	responsive	22.28	114.17	12708.84	251.82	252.09	252.71
66.35.20.123	responsive	37.34	-121.89	3026.80	62.20	80.61	140.74
66.35.22.79	responsive	38.90	-77.04	862.59	31.15	33.85	37.09
66.35.27.207	responsive	32.78	-96.81	1223.42	31.59	36.84	52.81
66.35.30.9	responsive	49.25	-123.12	2984.89	54.46	65.73	97.39
66.35.31.81	responsive	35.69	139.69	10303.47	141.65	144.18	148.44
69.168.185.43	responsive	51.05	-114.09	2396.80	44.37	51.58	71.93
69.48.236.198	responsive	18.52	73.86	13165.59	282.52	286.63	293.49
69.48.237.66	responsive	41.01	28.95	8887.60	144.88	150.84	169.41
69.48.238.200	responsive	25.08	55.31	11750.75	259.33	260.59	261.83
69.48.239.124	responsive	-26.20	28.04	13907.33	245.23	249.57	259.65
69.79.207.214	responsive	18.00	-76.79	2675.85	60.63	62.20	63.83
70.38.95.143	responsive	45.51	-73.59	1221.27	30.68	32.10	36.67
72.1.86.46	responsive	-36.85	174.76	13175.24	183.80	187.11	190.05
74.49.158.20	responsive	45.51	-73.59	1221.27	73.49	76.41	81.72
77.247.126.223	responsive	34.05	-118.24	2845.64	65.45	68.77	74.36
78.129.130.58	responsive	51.52	-0.72	6380.78	97.91	99.92	101.03
79.127.209.1	responsive	-33.46	-70.65	8383.28	153.58	160.79	180.80
80.67.167.93	responsive	48.85	2.35	6710.83	101.86	106.87	120.37
81.17.61.145	responsive	51.51	-0.13	6418.48	96.35	103.54	127.11
82.197.188.129	responsive	47.37	8.55	7190.80	113.76	115.40	118.15
84.17.41.11	responsive	47.61	-122.33	2914.04	56.90	59.72	62.81
84.17.57.129	responsive	22.28	114.17	12708.84	194.17	196.10	198.03
84.17.59.129	responsive	45.46	9.19	7348.35	117.55	118.78	120.59
87.249.137.8	responsive	53.33	-6.25	5957.00	111.22	114.19	122.99

Ping results and coordinates - 169-188

IP	Status	Lat	Lon	km	Min ms	Mean ms	Max ms
88.213.191.247	responsive	47.70	8.57	7172.05	114.03	115.59	117.91
88.218.137.64	responsive	44.80	20.47	8096.11	125.86	130.00	142.27
89.187.160.1	responsive	35.69	139.69	10303.47	157.31	162.45	180.82
89.187.162.1	responsive	1.29	103.85	15249.81	260.67	331.29	406.08
89.187.164.1	responsive	32.78	-96.81	1223.42	31.43	32.05	32.61
89.187.165.1	responsive	47.37	8.55	7190.80	130.03	136.13	152.62
91.199.118.184	responsive	50.12	8.68	7034.59	118.95	120.12	121.59
92.119.16.139	responsive	33.75	-84.39	775.41	18.94	24.28	35.45
93.177.80.27	responsive	59.33	18.07	6981.69	116.43	121.05	137.06
95.143.54.63	responsive	47.37	8.55	7190.80	112.66	117.72	134.48
95.211.20.80	responsive	52.37	4.89	6679.93	106.94	118.11	158.38
95.67.49.134	responsive	50.45	30.52	8221.87	147.78	152.67	168.29
96.45.38.22	responsive	1.29	103.85	15249.81	221.22	221.87	222.41
96.45.39.38	responsive	50.12	8.68	7034.59	115.53	117.97	120.36
96.45.40.45	responsive	51.51	-0.09	6420.39	95.80	101.22	107.35
96.45.41.167	responsive	43.63	7.00	7317.69	115.66	128.30	177.29
96.45.42.156	responsive	48.85	2.35	6710.83	104.33	119.53	153.32
96.45.43.6	responsive	43.71	-79.40	719.90	23.63	25.56	32.87
96.45.44.87	responsive	-33.87	151.21	14904.22	208.70	263.52	459.45
173.230.6.130	responsive	40.43	-86.91	0.00	4.81	11.19	32.39